

Ruthenium-106 versus iodine-125 plaque brachytherapy of 571 choroidal melanomas with a thickness of ≥ 5.5 mm

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ABSTRACT

Background Episcleral brachytherapy is the most common eye-preserving treatment for medium-sized choroidal melanomas. γ -emitting iodine-125 (^{125}I) and β -emitting ruthenium-106 (^{106}Ru) are widely used. The latter is however generally reserved for thinner tumours (<6 mm). In this study, we compare ocular and patient survival in thicker tumours treated with the respective radioisotope.

Methods All patients with ≥ 5.5 mm thick choroidal melanomas who were treated with plaque brachytherapy at a single institution between 1 November 1979 and 31 December 2015 were included ($n=571$). Size-controlled Cox regression HRs for postbrachytherapy enucleation, repeated brachytherapy and melanoma-related mortality were calculated, as well as Kaplan-Meier disease-specific survival and relative 10-year survival in matched subgroups.

Results 317 patients were treated with ^{106}Ru and 254 with ^{125}I . The rate of repeated brachytherapy was significantly higher among patients treated with ^{106}Ru (8%) than with ^{125}I (1%, $p<0.001$). Size-controlled Cox regression HRs for postbrachytherapy enucleation (^{125}I vs ^{106}Ru 0.7, $p=0.083$) and melanoma-related mortality were not significant (^{125}I vs ^{106}Ru 1.1, $p=0.63$). Similarly, Kaplan-Meier disease-specific and relative 10-year survival was comparable in matched groups of 5.5–7.4 mm (relative survival ^{106}Ru 59%, ^{125}I 56%) and ≥ 7.5 mm thick tumours (relative survival ^{106}Ru 46%, ^{125}I 44%).

Conclusions Rates of repeated brachytherapy were significantly higher among patients treated with ^{106}Ru versus ^{125}I for thick choroidal melanomas. There were, however, no significant differences in rates of enucleation or patient survival.

INTRODUCTION

Uveal melanoma is the most common primary intraocular malignancy in adults, with a high propensity for metastasis.¹ Historically, enucleation of the affected eye was the only treatment resort until the 1970s, when eye-conserving radiotherapy and brachytherapy started seeing increased use. In 2001, the Collaborative Ocular Melanoma Study demonstrated equal 5-year survival rates for enucleation and plaque brachytherapy with the γ -emitting radioisotope iodine-125 (^{125}I) of medium-sized tumours (2.5–10 mm in thickness and <15 mm in

diameter).² Local treatment failure, defined as intra-ocular tumour growth, recurrence or extraocular extension, had been observed in 10% of patients, and 13% had undergone postbrachytherapy enucleation.³ Greater tumour apical height and poorer baseline visual acuity were independent predictors for enucleation for reasons other than treatment failure. Later, the 10 years confirmed or suspected melanoma-related mortality was reported to be 21% and 22% in the enucleation and brachytherapy arms, respectively.⁴

Widely adopted in Europe, ruthenium-106 (^{106}Ru) is an alternative to ^{125}I .^{5–6} This β -emitting radioisotope has a threefold faster dose fall-off than ^{125}I , with increasingly lower relative energies delivered for each millimetre of tissue penetration (figure 1).^{7–9} This implies that for the same absorbed dose at the tumour apex, higher doses have to be given to the sclera and the base of the tumour. Beyond the tumour apex, the β -source delivers lower energies to surrounding tissues including the retina and the optic nerve.^{5,10,11} Previous reports also indicate that there is a greater lateral constriction of isodoses at the edge of a ^{106}Ru plaque, entailing a greater risk for recurrence with a narrow tumour-plaque margin.^{5,12} This steep dose fall-off has led to concerns that ^{106}Ru might be insufficient for thicker tumours, with increased risk for relapse and tumour death.^{5,13,14} Brachytherapy with ^{106}Ru is, therefore, generally used for tumours no thicker than 6–7 mm, even though some authors have shown promising results for tumours with apical thickness of ≥ 8.0 mm.¹¹ Tumours with a diameter of >16 mm carry a poor prognosis, have been excluded from most brachytherapy studies and are generally enucleated, regardless of what plaques and radioisotopes are available at individual institutions.

One previous study has reported enucleation, tumour recurrence and metastasis in 34, 13% and 55% of patients 10 years after brachytherapy of ≥ 8.0 mm thick tumours with one of ^{106}Ru , ^{125}I , cobalt-60 or iridium-192.¹⁵ In the same study, ^{106}Ru entailed a higher risk for tumour recurrence and postbrachytherapy enucleation than ^{125}I , but neither the difference in apex dose nor patient survival were compared between the radioisotopes. A third study found similar overall survival after ^{106}Ru and ^{125}I plaque brachytherapy of <5 mm thick uveal melanomas, but significantly



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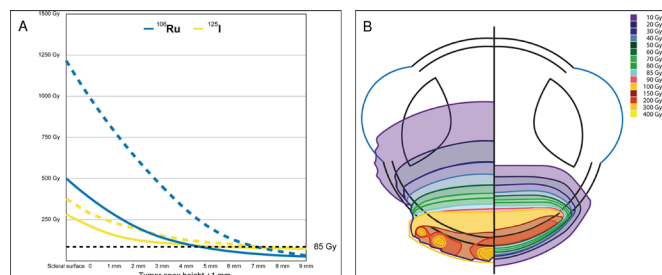


Figure 1 (A) Approximate dose fall-off curves for ruthenium-106 (^{106}Ru , blue) and iodine-125 (^{125}I , yellow) plaques when 85 Gy is prescribed at the apex +1 mm of 5 mm thick (solid lines) and 7 mm thick (dotted lines) tumours. For every millimetre of penetration into the tissue, the dose fall-off for the β -emitting ^{106}Ru is approximately threefold faster than for the γ -emitting ^{125}I . This implicates that scleral doses have to be significantly higher with ^{106}Ru than ^{125}I for equivalent doses at the tumour apex. (B) Comparison of theoretical two-dimensional dose distributions of a ^{125}I plaque (left) and a ^{106}Ru plaque (right), both planned with a dose of 85 Gy to the apex of a thin tumour. ^{125}I delivers higher doses to tissues beyond the tumour, and to tissues lateral to the plaque margin. B reprinted with permission from Eckert & Ziegler BEBIG, Berlin, Germany.

improved progression-free and enucleation-free survival with ^{106}Ru .¹⁶

In Sweden, brachytherapy with ^{106}Ru was introduced in 1979 and quickly became the principal treatment for tumours thinner than 7 mm. For thicker tumours, primary enucleation retained its mainstay role. In cases where the affected eye was the patient's last seeing eye or where the patient preferred globe-sparing therapy, tumours larger than 7 mm have been treated with ^{106}Ru . In 1999, ^{125}I was introduced and replaced enucleation as the treatment of choice for tumours with a thickness of 6–10 mm. Consequently, our records have a large number of patients with similar clinical presentations and tumour sizes that have been treated with either ^{106}Ru or ^{125}I due to historical changes in radioisotope use. We, therefore, see an opportunity to review our brachytherapy experience since 1979: In 571 consecutive patients with ≥ 5.5 mm thick tumours, we compare ^{106}Ru and ^{125}I brachytherapy side by side against local control and patient survival measurements.

METHODS

All patients who received ^{106}Ru or ^{125}I brachytherapy for choroidal melanoma at the Pathology and Oncology Service, St. Erik Eye Hospital from 1 November 1979 to 20 November 2017 were considered for the study ($n=1857$). Retrospective data were retrieved from digitalised clinical records and from the digital Brachytherapy of Uveal Melanoma directory (BRUM), including information on tumour size, treatment, retreatment, rates of transpupillary thermotherapy (TTT), enucleations and survival.

In review of the data in the BRUM directory, 295 patients were excluded as they had actually been enucleated rather than treated with plaque brachytherapy (indications: 274 large tumour size and/or location inaccessible for satisfactory plaque placement, 13 invasion of the optic disc, 3 patient preference, 1 uveitis, 1 chronic vitreous bleeding, 1 total retinal detachment, 1 extrascleral tumour growth and 1 suspected concurrent choroidal prostate cancer metastasis). Twelve patients were excluded as they did not have date of treatment specified and five because the site of origin of their tumour was the ciliary body or iris,

as opposed to the choroid. One patient was excluded since no measurement of the tumour's thickness was available. Additionally, all tumours treated after 31 December 2015 ($n=51$) were excluded to allow for a minimum of 2.5 years of follow-up data, yielding a total of 1493 patients eligible for an initial analysis.

Our protocol for ^{106}Ru brachytherapy allows for a maximum scleral dose of 1500 Gy.⁶ This enables a dose of 100 Gy at the prescription point of the apex of tumours up to 7 mm in thickness. +1 mm is added to adjust for the typical distance between the plaque and tumour base. For ^{125}I , the treatment was planned for 100 Gy to the apex +1 mm until the middle of 2003, after which it was lowered to 80 Gy due to concern for surrounding tissues with the deeper penetration of γ -radiation.

All ruthenium plaques were of the CCA, CCB, CCX, CCZ, COB, CIA or CIB types (Eckert & Ziegler BEBIG, Berlin, Germany), with diameters of up to 20 mm to allow for treatment of tumours with a diameter of up to 16 mm with 2 mm safety margin. Iodine plaques were custom made by a goldsmith in 18 carat gold alloy to replicate the ruthenium plaques in size and shape, with iodine seeds (Eckert & Ziegler) glued to the concave surface. Source specification data from the manufacturer were independently verified by medical physicists at the Karolinska University Hospital, Stockholm.

All patients had the tumour thickness and diameter measured preoperatively by standardised A and B scan ultrasonography. Regular follow-up was scheduled at 1, 3, 6 and 12 months after therapy, and then annually or semiannually if local control had been achieved. When tumour regression was deemed insufficient or there was tumour progression 6 months after brachytherapy or later, a second round of brachytherapy, TTT or enucleation could be considered. The main reason for such interventions could be found in the clinical record. Semiannual screening for liver metastases by ultrasonography or CT for 5 years after choroidal melanoma diagnosis was performed. After a few years, patients living outside the Stockholm area may be referred back to their home clinic. This does not affect data on repeated brachytherapy as these are exclusively performed at our institution. Similarly, data from enucleations will not be affected as all specimens are sent for histopathological examination at our institution.

The National Mortality registry is administered centrally and fed with data from all healthcare institutions in the country, including classification of cause of death. As there is a risk of misclassification of the actual cause of death, primarily between cutaneous and choroidal melanoma, we cross-checked the Mortality Registry's data with our own clinical records. If a patient had been treated for choroidal melanoma, developed one or several metastases of choroidal melanoma, and later died from metastasised melanoma, we considered the actual cause of death to be metastasised choroidal and not skin melanoma.

Statistical analysis

In power calculations, we considered the largest similar cohort published to date, in which the relative risk for tumour recurrence and enucleation after ^{106}Ru versus ^{125}I brachytherapy was 13.75 and 3.12, respectively.¹⁵ To detect the smallest of these differences (risk for enucleation) with 80% power and a level of significance (α) of 0.05, the sample size for each arm should be nine patients. Reversely, in our sample size of 317 patients treated with ^{106}Ru and 254 with ^{125}I , we would be able to detect differences of down to approximately 8% with 80% power and approximately 9% with 90% power. The Mann-Whitney U test was used for tests of the null hypothesis in non-parametrically

distributed group level data. The Pearson χ^2 test was used for comparisons of proportions in two-by-two tables. For analyses of hazard for postbrachytherapy enucleations, repeated brachytherapy and melanoma-related mortality, multivariate Cox regressions with tumour size variables as covariates were used. In reports of disease-specific and overall survival, Kaplan-Meier statistics were computed in matched subgroups and The log-rank (Mantel-Cox) test of equality of survival distributions applied. Groups were matched in consideration of recommendations for 1:1 nearest neighbour matching by exclusion of patients with extreme values in any of seven variables (age at diagnosis, sex, pretreatment visual acuity, tumour thickness, tumour diameter, American Joint Committee on Cancer (AJCC) T-category¹⁷ and follow-up years), until all differences became non-significant in Mann-Whitney U tests.¹⁸ Visual acuity was measured on a decimal scale chart at a distance of 5 m (Monoyer-Granström) ranging from 1.0 (corresponding to 20/20) to 0. Modified from standards in the Swedish National Quality Registry for Cataracts, counting fingers at a distance of 4 m was recorded as 0.08, hand movements as 0.04, perception and localisation of light as 0.01 and amaurosis as 0. Average visual acuities were calculated on -1 LogMar converted values, which were then converted back to the decimal scale. To allow for statistical comparison of mean AJCC T-category across matched subgroups, T categories were codified as T1=1, T2=2, T3=3 and T4=4. To test whether our data met the proportional hazard assumption, we built a Cox regression model with a time-dependent versus a time-independent treatment variable (^{106}Ru or ^{125}I brachytherapy). Disease-specific survival was defined as the time in years from brachytherapy to death from metastatic uveal melanoma. Patients were censored at death from other causes or at last occasion they were seen alive. This makes our reported follow-up time significantly shorter than the time from intervention to data collection, to better reflect our true observation time. Relative survival was defined as the observed cumulative proportion of patients surviving 10 years after brachytherapy divided by the expected cumulative proportion surviving in the general population according to year-per-year statistics for the general population.¹⁹ Differences with a $p < 0.05$ were considered significant, all p values being two sided. All statistical analyses were performed using IBM SPSS statistics V.25

RESULTS

Descriptive statistics

The 1493 patients had a distribution of tumour thicknesses that was left-skewed from the mean, with a high frequency of 2–7 mm thick tumours and a relatively lower frequency of thicker tumours (figure 2). Of these patients, 1233 were treated with ^{106}Ru and 260 with ^{125}I .

Of the 922 tumours that were <5.5 mm thick, 917 had been treated with ^{106}Ru and only 5 with ^{125}I .

For the remaining 571 patients, of whom 317 were treated with ^{106}Ru and 254 with ^{125}I , we found that the predictor satisfied the proportional hazard assumption in a Cox regression model with a time-dependent vs a time-independent treatment variable (^{106}Ru or ^{125}I brachytherapy, $p=0.78$). Patients treated with the respective radioisotope essentially matched each other for mean age at choroidal melanoma diagnosis. Although both ^{106}Ru and ^{125}I had been given to all tumours within the recommended dose limits of 70–100 Gy to apices, ^{106}Ru had been given in significantly higher doses, (mean apex +1 mm ^{106}Ru 91.6 Gy vs ^{125}I 81.4 Gy, Mann-Whitney U test $p < 0.001$) to significantly smaller tumours (^{106}Ru mean thickness 6.6 mm vs

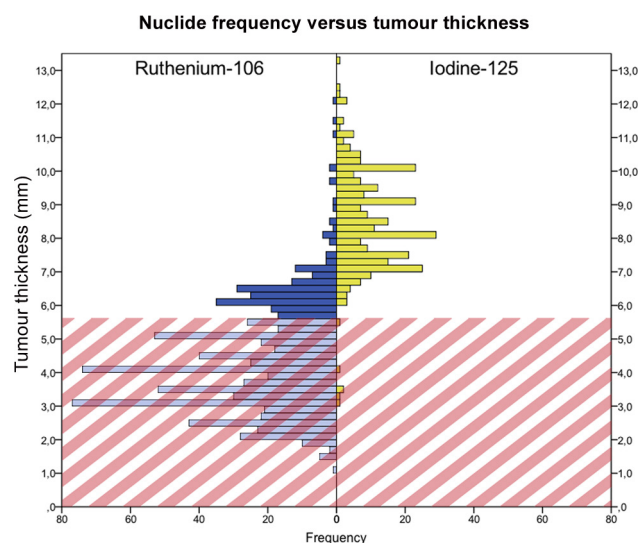


Figure 2 Tumour thickness frequency for patients treated with ^{106}Ru (left, blue) and ^{125}I (right, yellow). Of 922 excluded tumours that were <5.5 mm thick (striped area), 917 had been treated with ^{106}Ru and only 5 with ^{125}I . Additionally, all tumours treated after 31 December 2015 ($n=52$) were excluded to allow for a minimum of 2.5 years of follow-up data. Five hundred and seventy-one patients remained, of whom 317 were treated with ^{106}Ru and 254 with ^{125}I , iodine-125; ^{106}Ru , ruthenium-106.

^{125}I mean 8.5 mm, Mann-Whitney U test $p < 0.001$) with significantly better pretreatment visual acuity. Mean follow-up years were similar between the radioisotopes (mean 8.0 vs 7.9 years for patients treated with ^{106}Ru and ^{125}I , respectively, Mann-Whitney U test $p=0.77$).

Before 1 January 2000, all ($n=40$) of the thickest tumours (>7.4 mm) elected for brachytherapy were treated with ^{106}Ru , as no other radioisotope was available. After this date, 96% ($n=190$) of the thickest tumours were treated with ^{125}I and only 4% ($n=7$) with ^{106}Ru .

The majority of patients treated with TTT prior to or in adjunct to brachytherapy (148 of 170, 87%) were in the excluded group of <5.5 mm thick tumours, and no significant difference was seen between radioisotopes in ≥ 5.5 mm group (14 ^{106}Ru , 10 ^{125}I , χ^2 $p=0.87$). Neither were there any significant differences in the proportions of TTT after ^{106}Ru and ^{125}I (11 registered TTT distributed over all thicknesses, χ^2 $p=0.97$, table 1).

Rates of repeated brachytherapy

Data on rates of repeated brachytherapy were available from the year 2000. During the period, 14 of 181 patients (8%) that had undergone primary brachytherapy with ^{106}Ru and 3 out of 254 patients (1%) that had undergone primary brachytherapy with ^{125}I underwent a second round of brachytherapy (χ^2 $p < 0.001$). None of these had their first brachytherapy before the year 2000. Among patients treated with ^{106}Ru , the main reason for the repeated brachytherapy was tumour relapse in 12 and insufficient regression in 2. Among the three patients treated with ^{125}I , the main reason for the repeated brachytherapy was tumour relapse in all three. The mean interval from first to second brachytherapy was 2.2 years after ^{106}Ru (SD 1.7) and 1.25 years for ^{125}I (SD 0.9, Mann-Whitney U test $p=0.51$). Of the 17 patients who underwent repeated brachytherapy, eight eventually died from metastasised disease.

Table 1 Demographics and clinical features of patients included in this study

Plaque brachytherapy	Ruthenium-106	Iodine-125	P value*
n=	317	254	
Mean age at diagnosis, (range)	63 y (14–93)	62 y (18–89)	0.76
Pretreatment visual acuity, mean (range)	0.11 (0–1.3)	0.08 (0–1.3)	0.03
Tumour eye laterality, n (%)			<0.001
Right	101 (32)	121 (48)	
Left	80 (25)	133 (52)	
N/A	136 (43)	0 (0)	
Mean tumour thickness, mm (range)	6.6 (5.5–13.1)	8.5 (5.5–13.2)	<0.001
Mean tumour diameter, mm (range)	12.0 (5.0–23.0)	13.3 (6.0–21.0)	<0.001
Tumour thickness, n			0.27
5.5–6.4 mm	177	2	
6.5–7.4 mm	92	63	
7.5–8.4 mm	29	67	
8.5–9.4 mm	12	55	
>9.4 mm	7	67	
Intended radiation dose, mean (SD)			
Tumour apex	138.2 Gy (32.7)	102.5 (9.6)	<0.001
Tumour apex +1 mm	91.6 Gy (20.2)	81.4 Gy (7.6)	<0.001
Scleral surface	1057.5 Gy (298.9)	430.0 Gy (247.8)	<0.001
Treatment duration, mean (SD)			
Intended	194.9 hours (73.2)	80.7 hours (28.6)	<0.001
Actual	177.5 hours (54.6)	79.8 hours (28.5)	<0.001
Previous TTT†, n (%)	14 (4)	10 (4)	0.87
Follow-up‡, mean (SD, minimum-maximum)	8.0 y (3.9, 2.5–18.0)	7.9 y (3.5, 2.5–16.0)	0.77
Treatment year, n			0.23
1979–1985	11	0	
1986–1990	42	0	
1991–1995	35	0	
1996–2000	62	2	
2001–2005	59	57	
2006–2010	53	89	
2011–2015	55	106	

Demographics and clinical features of patients and tumours included in this study. Average visual acuity calculated after conversion to LogMAR equivalents. †, Proportion of patients that had been treated with transpupillary thermotherapy at any point prior to brachytherapy. ‡, Follow-up was terminated at time of death, or on last occasion patient was seen alive. *Mann-Whitney U-test for continuous variables, Pearson Chi-square test for two-by-two tables.

Gy, Gray; N/A, not available; SD, Standard deviation; y, Years.

Hazard for enucleation

Sixty-two of 317 patients (20%) underwent enucleation at any point after ^{106}Ru brachytherapy and 43 of 254 patients (17%) after ^{125}I brachytherapy (χ^2 $p=0.81$).

Among patients treated with ^{106}Ru , the main reason for the postbrachytherapy enucleation was tumour relapse in 55, total retinal detachment in 3, insufficient effect of brachytherapy and TTT in 2, pain and poor visual acuity in 1 and blindness without pain in 1.

Among patients treated with ^{125}I , the main reason for the postbrachytherapy enucleation was tumour relapse in 30, total retinal detachment in 2, chronic vitreous bleeding in 2, high intraocular pressure without diagnosis of iris rubeosis in 2, high intraocular pressure with iris rubeosis in 2, radiation retinopathy in 2, tumour necrosis filling the anterior chamber in 1, patient preference in 1 and chronic uveitis in 1.

In a Cox regression with tumour thicknesses, diameter and the respective radioisotope as covariates, the hazard for post-brachytherapy enucleation was increased with increasing tumour thickness (HR 1.3 for each increased thickness mm, 95% CI 1.2 to 1.5, $p<0.001$) but not significantly increased with increasing tumour diameter (HR 1.0 for each increased diameter mm, 95%

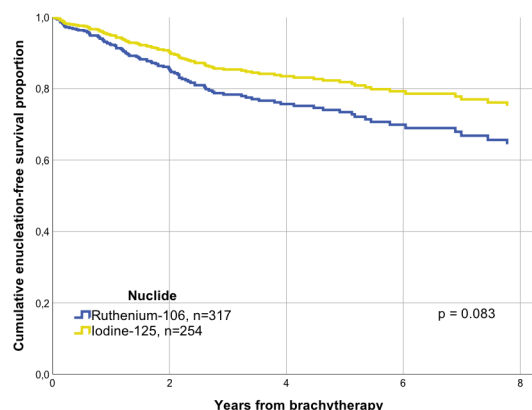


Figure 3 Bivariate Cox regression, enucleation-free survival at mean of tumour thickness and diameter for patients treated with ^{106}Ru (blue) and ^{125}I (yellow) brachytherapy. HR ^{125}I vs ^{106}Ru 0.65, 95% CI 0.4 to 1.1, $p=0.083$. ^{125}I , iodine-125; ^{106}Ru , ruthenium-106.

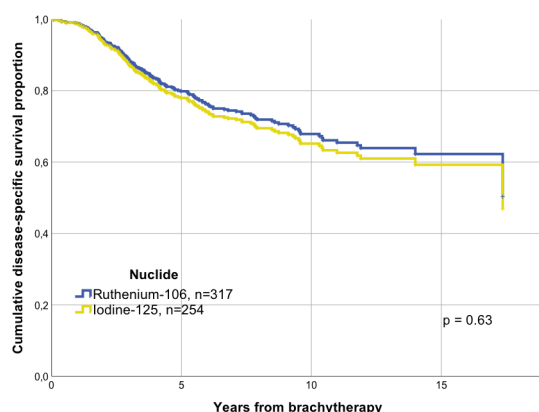


Figure 4 Bivariate Cox regression, disease-specific survival at mean of tumour thickness and diameter for patients treated with ^{106}Ru (blue) and ^{125}I (yellow) brachytherapy. HR ^{125}I vs ^{106}Ru 1.1, 95% CI 0.7 to 1.5, $p=0.63$. ^{125}I , iodine-125; ^{106}Ru , ruthenium-106.

CI 0.9 to 1.1, $p=0.71$) or with either radioisotope (HR ^{125}I vs ^{106}Ru 0.7, 95% CI 0.4 to 1.1, $p=0.083$, figure 3).

Patient survival

Including all 571 patients, there were no significant differences in hazard for melanoma-related mortality after ^{125}I and ^{106}Ru brachytherapy: In a multivariate Cox regression with tumour thicknesses, diameter and the respective radioisotope as covariates, the HR after ^{125}I vs ^{106}Ru brachytherapy was 1.1 (95% CI 0.7 to 1.5, $p=0.63$, figure 4). In bivariate regressions with either tumour thickness or diameter as covariates, HRs were 0.9 (95% CI 0.7 to 1.3, $p=0.61$) and 1.0 (95% CI 0.7 to 1.3, $p=0.89$), respectively. Individually, the hazard for

melanoma-related mortality was significantly increased at 1.2 for both each increased mm in tumour diameter (95% CI 1.1 to 1.2, $p<0.001$), and each increased mm in tumour thickness (95% CI 1.1 to 1.2, $p<0.001$).

In a subgroup analysis of 5.5–7.4 mm thick tumours, patients treated with ^{125}I still had significantly thicker tumours (mean thickness 7.0 mm, SD 0.3) than patients treated with ^{106}Ru (mean thickness 6.2 mm, SD 0.5, Mann-Whitney U test $p<0.001$). 55 of the patients treated with ^{106}Ru were, therefore, compared with 55 patients treated with ^{125}I , matched for patient age at diagnosis, sex, pretreatment visual acuity, mean tumour thickness, mean tumour diameter, AJCC T-category¹⁷ and follow-up years (table 2). In these 55+55 patients, 5 and 10 years melanoma-related mortality was 30% and 45% after ^{106}Ru brachytherapy, vs 23% and 38% after ^{125}I brachytherapy. Estimated Kaplan-Meier disease-specific median survival in these two matched groups was not significantly different at 13.3 years after ^{106}Ru brachytherapy (95% CI 11.3 to 15.3) vs 11.5 years after ^{125}I brachytherapy (95% CI 10.3 to 12.8, log-rank $p=0.26$, figure 5A).

In tumours with a thickness of ≥ 7.5 mm, patients treated with ^{125}I had significantly thicker tumours (mean thickness 9.0 mm, SD 1.2) than patients treated with ^{106}Ru (mean thickness 8.6 mm, SD 1.1, Mann-Whitney U test $p=0.003$). In analogy with the smaller group, 45 ≥ 7.5 mm tumours treated with ^{125}I were, therefore, compared with 45 matched tumours treated with ^{106}Ru (table 2). In these 45+45 patients, 5 and 10 years melanoma-related mortality was 25% and 51% after ^{106}Ru brachytherapy, vs 33% and 54% after ^{125}I brachytherapy. Estimated Kaplan-Meier disease-specific median survival was not significantly different at 9.7 years after ^{106}Ru brachytherapy (95% CI 3.1 to 16.4) vs 9.8 years after ^{125}I brachytherapy (95% CI 6.3 to 13.4, log-rank $p=0.89$, figure 5B).

Table 2 Baseline characteristics in matched cohorts

	Ruthenium-106	Iodine-125	P value*
5.5–7.4 mm thick tumours			
n	55	55	
Mean age at diagnosis, (range)	63 y (14–93)	63 y (18–89)	0.18
Females/males, n	23/32	23/32	1.0
Pretreatment visual acuity, mean (range)	0.12 (0–1.3)	0.12 (0–1.3)	0.64
Mean tumour thickness, mm (range)	6.9 (6.4–7.4)	6.9 (6.0–7.4)	0.64
Mean tumour diameter, mm (range)	12.3 (6.0–18.0)	12.3 (6.0–18.0)	0.82
AJCC T-category (mean, mode, range)	T2, T2, T1–T3	T2, T2, T1–T3	0.32
Follow-up, mean (SD, minimum-maximum)	6.2 y (3.9, 2.5–16.5)	6.9 y (3.5, 2.5–11.0)	0.19
Mean tumour apex dose, Gy (SD, minimum-maximum)	91 (13, 58–103)	81 (4, 79–100)	<0.001
Mean scleral dose, Gy (SD, minimum-maximum)	1027 (553, 265–1494)	285 (150, 200–583)	<0.001
≥ 7.5 mm thick tumours			
n=	45	45	
Mean age at diagnosis, (range)	64 y (32–86)	65 y (43–86)	0.99
Females/males, n	22/23	21/24	0.83
Pretreatment visual acuity, mean (range)	0.20 (0–1.3)	0.12 (0–1.3)	0.55
Mean tumour thickness, mm (range)	8.5 (7.5–11.6)	8.5 (7.5–12.2)	0.74
Mean tumour diameter, mm (range)	12.1 (5.0–23.0)	12.3 (7.0–19.0)	0.71
AJCC T-category (mean, mode, range)	T3, T3, T2–T4	T3, T3, T2–T4	0.10
Follow-up, mean (SD, minimum-maximum)	6.0 y (1.6, 2.5–17.0)	6.8 y (2.1, 2.5–18.0)	0.48
Mean tumour apex dose, Gy (SD, minimum-maximum)	74 (18, 45–100)	80 (3, 80–100)	0.03
Mean scleral dose, Gy (SD, minimum-maximum)	1212 (392, 310–2106)	384 (490, 224–870)	<0.001

Comparison of baseline characteristics in matched cohorts of 5.5–7.4 mm thick tumours (top) and ≥ 7.5 mm thick tumours (bottom). Average visual acuity calculated after conversion to LogMAR equivalents. *p-values as determined by Mann-Whitney U-tests, except female/male ratios, which were determined by Pearson Chi-square tests. Gy, Gray; SD, Standard deviation; y, Years.

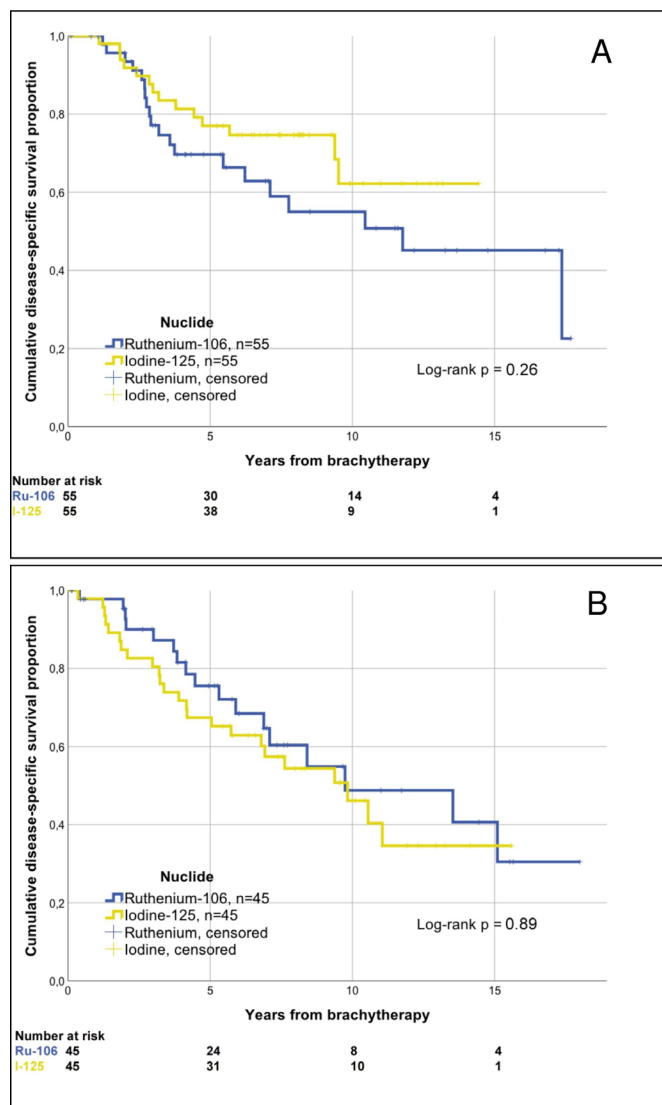


Figure 5 Kaplan-Meier curves, estimated disease-specific survival after brachytherapy with ^{106}Ru and ^{125}I within separate tumour thickness intervals. (A) Fifty-five tumours treated with ^{106}Ru within the 5.5–7.4 mm thickness interval compared with 55 tumours treated with ^{125}I , matched for patient age at diagnosis, sex, pretreatment visual acuity, mean tumour thickness, mean tumour diameter, AJCC T-category and follow-up years. Kaplan-Meier disease-specific median survival between the two radioisotopes was not significantly different at 13.3 years after ^{106}Ru brachytherapy (95% CI 11.3 to 15.3) vs 11.5 years after ^{125}I brachytherapy (95% CI 10.3 to 12.8 log-rank $p=0.26$). (B) Forty-five ≥ 7.5 mm tumours treated with ^{106}Ru compared with 45 tumours treated with ^{125}I , matched for patient age at diagnosis, sex, pretreatment visual acuity, mean tumour thickness, mean tumour diameter, AJCC T-category and follow-up years. Again, Kaplan-Meier disease-specific median survival between the two radioisotopes was not significantly different at 9.7 years after ^{106}Ru brachytherapy (95% CI 3.1 to 16.4) vs 9.8 years after ^{125}I brachytherapy (95% CI 6.3 to 13.4, log-rank $p=0.89$). ^{125}I , iodine-125; ^{106}Ru , ruthenium-106.

Relative survival

In the subgroup of matched 5.5–7.4 mm thick tumours ($n=110$), the 10-year relative survival rate after ^{106}Ru and ^{125}I brachytherapy was 59% and 56%, respectively.

In the equally matched subgroup of ≥ 7.5 mm thick tumours ($n=90$), the 10-year relative survival rate after ^{106}Ru and ^{125}I brachytherapy was 46% and 44%, respectively.

DISCUSSION

In summary, we found no significant or consistent differences in outcomes after ^{106}Ru - and ^{125}I brachytherapy of ≥ 5.5 mm thick choroidal melanomas with regard to size-controlled hazards for postbrachytherapy enucleation or melanoma-related mortality. Similarly, there were no significant differences in disease-specific survival, in melanoma-related mortality or in relative survival within matched subgroups. The exception was rates of repeated brachytherapy, which was significantly higher among patients treated with ^{106}Ru . This is noteworthy, not least since each brachytherapy session can cause distress to the patient and an extra burden for healthcare providers. Even though the reason for each instance of a repeated brachytherapy was stated in the medical records, the reason for the difference between the radioisotopes cannot be inferred from our data. One can perhaps assume that the faster fall-off with ^{106}Ru make greater demands on the precision of plaque placing over the tumour, with slightly decentred plaques leading to insufficient tumour margin doses and thereby the need for a repeated brachytherapy. Importantly, the increased rates of repeated brachytherapy with ^{106}Ru did not affect any of the other ocular or patient survival measurements.

Limitations to this study include those associated with any directory-based research: Our findings rely to a very large extent on the accuracy of data submitted by ourselves and other clinicians. The database might not capture all metastases and might misclassify melanoma-related deaths. We sought to limit the impact of such missing data and misclassifications by cross-reference to our own clinical records. Further, as our patients were not randomised to receive ^{106}Ru or ^{125}I brachytherapy, a selection bias during the period when both radioisotopes were available cannot be excluded. However, the fact that the vast majority (96%) of our thickest tumours were treated with ^{125}I after it became available indicates that the influence of other prognostic factors on the selection of radioisotope, that is, tumour location, was limited. We sought to further limit the impact of such factors by excluding melanomas originating in the iris and ciliary body, and by a subgroup analysis in which patients were matched by multiple factors. Third, even though it remains to be shown that lower doses entail a higher recurrence rate and worse survival,²⁰ it should be mentioned that ^{106}Ru was prescribed in significantly higher doses than ^{125}I during a period of the study, which may have influenced our results.

Overall, 18% of our patients underwent secondary enucleation in the first 5 years after brachytherapy. This is higher than the 5-year cumulative risk of enucleation of 12.5% after ^{125}I brachytherapy reported in the Collaborative Ocular Melanoma Study,³ but lower than rates reported by Lommatzsch *et al*,²¹ Shields *et al*¹⁵ and Mishra *et al*.²²

In conclusion, the grounds for strictly limiting the use of ^{106}Ru to tumours with a thickness of less than 6–7 mm may be questioned. Prospective and randomised studies would be required to draw more definitive conclusions regarding the superiority of either radioisotope.

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Competing interests None declared.

Patient consent for publication Not required.

Ethics approval The study adhered to the tenets of the Declaration of Helsinki and the protocol was approved by the regional ethical review board in Stockholm.

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